

JOSHUA COWELL

Lynchburg, VA | joshuacowell2005@gmail.com

EDUCATION

Bachelor of Science in Mechanical Engineering

Expected May 2027

Liberty University, Lynchburg, VA

GPA: 3.82

Relevant Coursework: Fluid Dynamics, Heat Transfer, Thermodynamics I & II, Differential Equations, Dynamics, Materials Engineering, Statics, Calculus I–III

PROFESSIONAL EXPERIENCE

Thermal Hydraulics Engineering Intern

May 2025 – Dec 2025

Framatome Nuclear

- Developed a predictive wear mathematical model in Python for critical reactor components, achieving **75% greater accuracy** compared to the previous model
- Programmed a Python algorithm using 6-dimensional Xarray to calculate safe reactor operating conditions, reducing computation time from **800+ hours to minutes**
- Performed system-level flow and vibration analysis using AFT Fathom and custom FIV modeling software
- Awarded **\$2,500** for final summer presentation; hired part-time during academic year

Chief Systems Engineer

June 2025 – Present

Liberty Rocketry (9th of 156 international teams, 2024)

- Lead member captain overseeing design, testing, and analysis of high-powered rocket systems
- Direct all engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems
- Coordinate communication between internal teams, external vendors, and university faculty
- Manage over \$16,000 in project budgeting and resource allocation across competition campaigns
- Utilize SolidWorks, ANSYS, and MATLAB to verify accuracy of subsystem models and simulations

Lead Undergraduate Research Assistant

Sep 2024 – Present

Liberty University

- Lead team of 5 undergraduate students performing split-Hopkinson pressure bar (SHPB) testing including sample preparation, high-speed imaging, and electron microscopy
- Design and manufacture custom SHPB systems tailored to specific material testing requirements

Senior Structural Engineer

Sep 2023 – June 2025

Liberty Rocketry

- Conducted simulations, FEA, and computational modeling using ANSYS Fluent, ANSYS Mechanical, RocketPy, OpenRocket, and Onshape
- Delivered technical reports ensuring rocket compliance with competition standards and flight requirements

Camp Counselor

May 2024 – Aug 2024

Woodlands Camp, GA

- Provided 24-hour supervision during weeklong outdoor expeditions with 12 youth (ages 8–18), instructing survival skills, rock climbing, whitewater rafting, and cave camping

TECHNICAL SKILLS

Software: AFT Fathom, ANSYS Fluent, ANSYS Mechanical, SolidWorks, Onshape, OpenRocket, RocketPy

Equipment: Scanning Electron Microscope (SEM), Metal X 3D Printer, CNC, WaterJet, Lathe

Programming: Python, MATLAB

PROJECTS

Predictive Wear Model

- Developed Python-based predictive wear model for reactor components, improving accuracy by 75% and enabling predictive maintenance strategies

Reactor Operating Limit Algorithm

- Engineered Python automation framework reducing 800+ hours of manual calculations to minutes with automated report generation

Heat Exchanger CFD Analysis

- Optimized heat exchanger design using ANSYS Fluent CFD simulations, generating flow and thermal visualizations

Unmanned Aerial Vehicle (UAV)

- Engineered custom UAV with 3D-printed components and validated aerodynamics via ANSYS CFD simulations

High Strain Rate Bone Research

- Investigating dynamic loading behavior of human bone using Split Hopkinson Pressure Bar for conference presentation